Aveen Dayal

AI Research Scientist | Multimodal and Generative AI | Efficient Inference and Model Evaluation

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Professional Summary

AI researcher with a Ph.D. in Artificial Intelligence and industry research experience at Dolby, Microsoft Research, and Adobe Research. Expertise in multimodal and generative AI, efficient inference, OOD robustness, content protection and attribution, and model evaluation. Published at CVPR, WACV, ECCV, NeurIPS, and IEEE TIP, with experience building and benchmarking LLM, multimodal, diffusion, and real-world vision prototypes.

Technical Skills

Programming and ML: Python, PyTorch, TensorFlow, scikit-learn, NumPy, OpenCV
Models and Methods: Hugging Face Transformers, multimodal models, diffusion models, parameter-efficient fine-tuning, prompt optimization
Systems and Evaluation: Distributed training/inference, profiling, APIs and model serving, Linux, Git, experiment tracking, OOD and human-aligned evaluation

Research and Industry Experience

Senior Multimodal AI Researcher — **Sep. 2025 – Present**
Dolby Advanced Technology Group India — *Bengaluru, India*

- Developing media-level watermarking methods for piracy deterrence in live-streaming content across offline and online settings; contributing to an invention currently being prepared for patent filing.
- Designed and executed a content-attribution study for text-to-image diffusion models, evaluating SAM3, DINO, and CLIP-based approaches on Stable Diffusion 1.5 and Stable Diffusion XL using precision and recall.
- Building a robust evaluation framework for generated multimodal content across **5 open-source models** and **20+ evaluation dimensions**, using benchmarks with human-alignment annotations to ground and validate model-generated scores.

Research Intern — **Jun. 2025 – Aug. 2025**
Adobe Research India — *Bengaluru, India*

- Built a proof-of-concept **layer-wise text-to-poster generation and editing pipeline** that used a multimodal LLM to reason over input text and existing poster elements before applying targeted modifications.
- Delivered an interactive demo and conducted side-by-side qualitative comparisons with GPT-4o. On the evaluated poster examples, the prototype showed stronger instruction adherence and fewer unintended edits.

Research Intern — **Jul. 2024 – Dec. 2024**
Microsoft Research India — *Bengaluru, India*

- Built and benchmarked a **tree-based speculative-decoding** method for **Vicuna-7B** and **Llama-3-Instruct-8B** on an NVIDIA A100 80GB GPU, achieving speedups of 2.91× and 2.62×, with 4.08 and 4.12 accepted tokens per target-model forward pass, respectively.
- Matched with benchmark EAGLE's Vicuna-7B speedup (2.91× vs. 2.90×) and improved accepted length over EAGLE on Vicuna-7B (4.08 vs. 3.94) and Llama-3-Instruct-8B (4.12 vs. 3.65), while characterizing throughput–acceptance trade-offs against EAGLE and EAGLE-2.
- Designed an **OOD evaluation** study for **Phi-3.5-Vision and LLaVA-OneVision** across 8 domains of OfficeHome and TerraIncognita datasets. Quantified in-distribution/OOD degradation and evaluated prompt optimization and adversarial LoRA strategies, identifying limitations of the tested interventions under domain shift.

Research Intern — **Mar. 2023 – Nov. 2023**
Autonomous and Cyber-Physical Systems Lab, University of Agder — *Grimstad, Norway*

- Developed adversarial-learning methods for domain generalization and multi-source domain adaptation, including theoretical analysis using unseen-target error bounds; the work resulted in first-author publications at **NeurIPS 2023** and **IEEE TIP 2025**.

Visiting Researcher — **Jan. 2020 – Jul. 2021**
Autonomous and Cyber-Physical Systems Lab, University of Agder — *Grimstad, Norway*

- Contributed to **six published interdisciplinary deep-learning projects** spanning biomedical imaging, sensor-based systems, and autonomous vehicles.
- Optimized and profiled lightweight models for **edge deployment** on NVIDIA Jetson AGX Xavier and Raspberry Pi 4B using TensorFlow Lite and ONNX, focusing on inference latency and memory footprint.
- Developed a real-time surround-view ADAS prototype that fused camera imagery and 3D point clouds for blind-spot reduction and object-distance estimation.

Education

Indian Institute of Technology Hyderabad — **Aug. 2021 – Dec. 2025**
Ph.D. in Artificial Intelligence, GPA: 9.19/10 — Hyderabad, India
Prime Minister's Research Fellow (PMRF)
Excellence in Research Award, 2024

BML Munjal University — **May 2020**
B.Tech. in Computer Science and Engineering, GPA: 9.44/10 — Gurgaon, India

Selected Publications

- **A. Dayal**, P. Divya, N. Tiwari, L. R. Cenkeramaddi, C. K. Mohan, and A. Kumar, "Bridging the domain gap in small multimodal models: A dual-level alignment perspective," ***IEEE/CVF WACV***, 2026.
- S. VCR, R. Lalla, **A. Dayal**, T. Kulkarni, A. Lalla, V. N. Balasubramanian, and M. H. Khan, "Foundation model priors enhance object focus in feature space for source-free object detection," ***IEEE/CVF CVPR***, 2026.
- **A. Dayal**, Shrusti S., L. R. Cenkeramaddi, C. K. Mohan, and A. Kumar, "Leveraging mixture alignment for multi-source domain adaptation," ***IEEE Transactions on Image Processing***, 2025.
- **A. Dayal**, R. Lalla, L. R. Cenkeramaddi, C. K. Mohan, A. Kumar, and V. N. Balasubramanian, "Improving unsupervised domain adaptation: A pseudo-candidate set approach," ***ECCV***, 2024.
- **A. Dayal**, V. K. B., L. R. Cenkeramaddi, C. K. Mohan, A. Kumar, and V. N. Balasubramanian, "MADG: Margin-based adversarial learning for domain generalization," ***NeurIPS***, 2023.
- **A. Dayal**, M. Aishwarya, S. Abhilash, C. K. Mohan, A. Kumar, and L. R. Cenkeramaddi, "Adversarial unsupervised domain adaptation for hand gesture recognition using thermal images," ***IEEE Sensors Journal***, 2023.
- **A. Dayal**, L. R. Cenkeramaddi, and A. Jha, "Reward criteria impact on the performance of reinforcement learning agent for autonomous navigation," ***Applied Soft Computing***, 2022.
- **A. Dayal**, S. R. Yeduri, B. H. Koduru, R. K. Jaiswal, J. Soumya, M. B. Srinivas, O. J. Pandey, and L. R. Cenkeramaddi, "Lightweight deep convolution neural network for background sound classification in speech signals," ***Journal of the Acoustical Society of America***, 2022.

Selected Applied AI Projects

Computer-Vision System for Aircraft Movement in Hangars — *Industry collaboration with Innominds*

- Co-developed a proof of concept that fused DeepLabV3 semantic segmentation from RGB cameras with LiDAR point clouds and implemented collision-avoidance logic to assist aircraft movement inside hangars.

Autonomous Drone-Based Medical Delivery System — *IIT Hyderabad BUILD initiative*

- Co-developed a DJI Tello navigation pipeline using its onboard camera feed to detect spatial QR-code destinations and execute autonomous landing for a medical-delivery proof of concept.

Real-Time Surround-View System for ADAS — *Bachelor's thesis / University of Agder*

- Built and deployed a 360-degree surround-view and object-distance estimation system on a TurtleBot3 by fusing 2D camera imagery with 3D point clouds for blind-spot reduction and collision avoidance.

Multi-Object Detection and Color Identification — *End-to-end IoT prototype*

- Developed an assistive-vision device using YOLOv2 and K-means clustering, achieving **76.8 mAP** for object detection and **85.2% accuracy** for color recognition.

Selected Honors and Recognition

- Awarded the **Excellence in Research Award 2024** by IIT Hyderabad; selected as one of 28 students across 15 departments.
- Awarded the **Prime Minister's Research Fellowship (PMRF)** in 2022.
- Received the **Microsoft Research Travel Grant** to attend NeurIPS 2023 in the United States and to attend ECCV 2024 in Italy.
- Invited to present research at the **3rd INMOST Workshop** on Indo-Norwegian Collaboration in Intelligent Offshore Mechatronics Systems, University of Agder, Grimstad, Norway.
- Invited to deliver multiple **talks on the Foundations of Machine Learning** at several institutes, Vardhaman College of Engineering, B.V. Raju Institute of Technology and Bharat Institute of Engineering and Technology.

Academic Service

Served as a reviewer for **CVPR, ICCV, ECCV, NeurIPS, ICML, and ICLR**, as well as several reputed journals.